Improving Precision & Power in Randomized Trials Using Covariate Adjustment

JSM 2026: CE 30 - Covariate Adjustment

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Recap & Notation

High-Level Recap on Covariate Adjustment

- Prognostic factors X; Measured before treatment assignment A;
- **Variance of Chance Imbalances** $\rightarrow$ Variance of unadjusted estiamates
  - Depends on Variance in Population; Inclusion/Exclusion; Design
  - $I = (\sum_{i=1}^{N} A_i X_i / \sum_{i=1}^{N} A_i) - (\sum_{i=1}^{N} (1 - A_i) X_i / \sum_{i=1}^{N} (1 - A_i))$
  - $\uparrow var(I) \rightarrow \uparrow var(\hat{\theta}_U)$ - Unadjusted estimate
- Design: stratified randomization - limits potential imbalance
- Analysis: Covariate Adjustment - Complement to design
  - Conditional (*model-based*): Correctly specified - *a priori*
  - Marginal (*model-assisted*): Working model not assumed to be correctly specified

Marginal Approach:

1. Estimate same quantity (estimand) as unadjusted analysis: θ
2. Same or weaker assumptions
3. Covariates are *prognostic*: $\uparrow$ Precision $\uparrow$ Power

- Adjustment improves precision by addressing chance imbalances in X using *working models* that aren't assumed to be correctly specified. Models used for prediction, not interpretation.
- Randomization $\rightarrow A \perp X \rightarrow$ robustness to misspecification.
  - Not guaranteed: Observational; Causal Inference methods
- Benefit: $var(I)$ - Gain *even if allocation is balanced*
  - Difference: $\hat{\theta}_A - \hat{\theta}_U$ depends on observed imbalance

Terminology, Notation, and Assumptions

- Observed outcome Y, treatment assignment A, baseline covariates X
  - $A = 1$: Treatment; $A = 0$: Control
- N Participants: $(X_i, A_i, Y_i) \overset{\text{iid}}{\sim} P$ where P is unknown
- Treatment assigned independently of X: $A \perp X$ in RCT
  - Stratified Randomization: $X = (X_S, X_{\bar{S}})$ then $A \perp\!\!\!\perp X_{\bar{S}} | X_S$
- Same/similar assumptions needed for consistency of unadjusted estimator
- Consistency of estimator: $\hat{\theta} \xrightarrow{p} \theta$ i.e. converges asymptotically to true value of the estimand

Continuous Outcome: (Bannick, Yi, and Ye 2026)

- Adjusted analyses: Unadjusted Estimate + Augmentation
  - $\hat{\theta}_A = \boxed{\hat{\theta}_U} + \boxed{h(X, A)}$: choose $h(X, A)$ to reduce $var(\hat{\theta}_A)$
- ANCOVA: Single Model - No Treatment Interactions
  - $\hat{\theta}_A = \boxed{\bar{Y}_1 - \bar{Y}_0} - \boxed{\hat{\beta}'(\bar{X}_1 - \bar{X}_0)}$
    - $\bar{X}_A$: Covariate Vector Mean in arm $A = a$
- ANHECOVA: Treatment-Stratified Models (All Interactions)
  - $\hat{\theta}_A = \boxed{\bar{Y}_1 - \bar{Y}_0} - \boxed{\left(\hat{\beta}_1{}'(\bar{X}_1 - \bar{X}_\bullet) - \hat{\beta}_0'(\bar{X}_0 - \bar{X}_\bullet)\right)}$
    - $\bar{X}_\bullet$: Covariate Vector Mean pooled across arms

Effect of Adjustment

- $(\widehat{\theta}_A - \widehat{\theta}_U)$: Difference Adjusted vs. Unadjusted
  - ANCOVA: $\boxed{\widehat{\beta}}' \boxed{(\bar{X}_1 - \bar{X}_0)}$
  - Association with outcome **and** Imbalance between arms
  - When X are relatively balanced: $\widehat{\theta}_U \approx \widehat{\theta}_A$
- Adjustment is reducing the variance in $\widehat{\theta}_U$ that is explained by imbalance in X.
  - Reduction depends on the **association with the outcome** and the **variance of the imbalance** but **not the observed value of the imbalance** (Wang, Ogburn, and Rosenblum 2019)
  - Reduced variance **even if observed allocation is relatively balanced.**
- Model robustness comes from randomization: In truth $A \perp X$

Hands-On

Learning Objectives: Hands-On

- Identify packages for adjusted analyses
- Identify resources for self-learning: tutorials, data
- Compute adjusted analyses for continuous, binary, ordinal, time-to-event outcomes
- Be able to estimate precision gain: historical data

Hands-On

- Focus on R: (+) Free & Open Source; (+) Capabilities; (-) Syntax
  - Stata: `teffects`; SAS Macros: DIY
  - New to R? See Learning & Using R section.
- What > How: Software generally fairly straightforward
  - Regression: Outcome Y, Treatment assignment A, Covariates X
    - Estimand, Design (Scheme, Strata), Variance Estimation
  - Lists of variable names (SAS, Stata, R) vs. Formulas (R)

Using R

- Download R and R Studio
    - Windows: RTools - Compilers; Built into iOS/Linux; **Only when needed.**
        - Building packages from source code vs. pre-compiled binary files: Source may have newer version than binaries.
- Download the .html - Easier to follow the code
    - Code can be copied by clicking upper right corner
- Download the .qmd - Run chunk-by-chunk
- Install packages: `install.packages` (CRAN) `pak::pak()` (GitHub)

Software:

- Several R packages on CRAN and GitHub (GH): Outcomes, Estimands

Package	Continuous	Binary	Ordinal	Time-to-Event	Repo
adjrct			✓	✓	GH
AIPW	✓	✓		✓	CRAN
drord			✓		GH
PSW	✓	✓			CRAN
RobinCar	✓	✓		✓	CRAN
RobinCar2	✓	✓		✓	CRAN
speff2trial	✓	✓		✓	CRAN
survtmle				✓	GH

- Need for software that meets standards of regulatory guidance
  - Validation plan; Testing workflow; Code coverage; Repository state

Example RCT Datasets

- `examplercts` package: 9 analysis-ready datasets: Dictionaries, Variable/Value Labels

Dataset	Continuous	Binary	Ordinal	Time-to-Event	Longitudinal
strep_tb		✓	✓		
indo_rct		✓			
botox_dystonia	✓				✓
laryngoscope	✓	✓	✓	✓	
licorice_gargle	✓	✓	✓		✓
supraclavicular	✓	✓	✓	✓	
perio_pregnancy	✓	✓	✓		✓
colon_cancer				✓	
lung_cancer				✓	

- RCT Bench - 125 completed trials: Searchable, Dictionaries (Shao et al. 2026)

CovariateAdjustment.GitHub.io

- Example Data & Code:
  - Tabulations, figures, working with R
- Estimands and estimation methods
- Survey of literature; Practical guidance
- Major update: Next two months
  - Missing Data; Double Robustness; Ordinal, Longitudinal analyses
- Future: Cluster RCTs, Information Monitoring, New methods

Installing Packages

```r
install.packages(
  pkgs = c("dplyr", "drord", "cobalt", "kableExtra", "lmtest",
           "pak", "sandwich", "speff2trial", "table1")
)

# Installing packages from GitHub
pak::pak(
  pkg =
    c("covariateadjustment/examplercts",
      "jbetz-jhu/drwls",
      "nt-williams/simul",
      "nt-williams/adjrct")
)
```

Load Packages

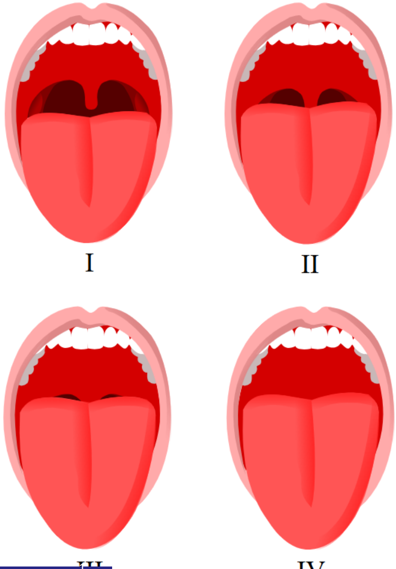
```
# Continuous, Binary Analyses
library(dplyr) # Wrangling data
library(drwls) # G-computation & DR-WLS
library(cobalt) # Standardized Differences
library(examplercts) # Example Datasets
library(kableExtra) # Printing Tables in HTML/TeX
library(lmtest) # Computing CIs with Robust SEs
library(sandwich) # Computing Robust SEs for GLMs
library(table1) # Tabulations

# Ordinal, Time-to-Event Analyses
library(adjrct) # Time-to-Event: Survival & RMST
library(drord) # Adjusted Ordinal Analyses
library(speff2trial) # Adjusted Marginal Hazard Ratio
library(survival) # Logrank tests; Cox PH Model
```

licorice_gargle Data in examplercts Package

- Licorice gargle (Treatment) vs. Sucrose Gargle (Control)
- Outcomes: Cough, Throat Pain at 30m, 90m, 4h post-extubation
    - Pain: 0 (None) - 10 (Worst Imaginable)
        - Treat as continuous and binary (0 vs. 1-10); Ordinal
    - Cough: No Cough < Mild < Moderate < Severe
- Covariates: Age, Sex, BMI, Physical Status, Mallampati Score

Mallampati Score: Source: Wikipedia



Tabulations & Balance:

- `table1::table1()` - Producing HTML/LaTeX summary tables

```
table1::table1(
  x = ~ age_bl + gender + asa_physical_status_bl +
    bmi_bl + mallampati_bl + pain_yn_bl | arm,
  data = licorice_gargle
)
```

- `cobalt::bal.tab` - Calculating Standardized Differences

```
cobalt::bal.tab(
  x =
    licorice_gargle %>%
    dplyr::select(dplyr::ends_with(match = "_bl")),
  treat = licorice_gargle$arm,
  binary = "std",
  continuous = "std",
  s.d.denom = "pooled"
)
```

Baseline Covariates

- Missingness; Low Frequency Categories; Balance

	0. 5g Sugar (N=117)	1. 0.5g Licorice (N=118)	Overall (N=235)
Age at Baseline			
Mean (SD)	58.0 (16.1)	56.7 (14.9)	57.4 (15.5)
Median [Min, Max]	63.0 [18.0, 86.0]	60.5 [19.0, 83.0]	62.0 [18.0, 86.0]
Gender			
0. Female	44 (37.6%)	49 (41.5%)	93 (39.6%)
1. Male	73 (62.4%)	69 (58.5%)	142 (60.4%)
ASA Physical Status			
1. Healthy	19 (16.2%)	22 (18.6%)	41 (17.4%)
2. Mild Systemic Disease	67 (57.3%)	67 (56.8%)	134 (57.0%)
3. Severe Systemic Disease	31 (26.5%)	29 (24.6%)	60 (25.5%)
Body Mass Index (kg/m^2)			
Mean (SD)	25.6 (4.25)	25.6 (4.32)	25.6 (4.27)
Median [Min, Max]	26.1 [15.6, 34.1]	25.7 [16.4, 36.3]	25.9 [15.6, 36.3]
Mallampati Score (BL)			
1. 1	31 (26.5%)	39 (33.1%)	70 (29.8%)
2. 2	69 (59.0%)	66 (55.9%)	135 (57.4%)
3. 3+	17 (14.5%)	13 (11.0%)	30 (12.8%)
Preoperative Pain (BL)			
0. No	115 (98.3%)	118 (100%)	233 (99.1%)
1. Yes	2 (1.7%)	0 (0%)	2 (0.9%)

Standardized Differences: Imbalance

```
## Balance Measures
##                                                          Type Diff.Un
## age_bl                                                Contin. -0.0854
## asa_physical_status_bl_1. Healthy                      Binary  0.0634
## asa_physical_status_bl_2. Mild Systemic Disease        Binary -0.0098
## asa_physical_status_bl_3. Severe Systemic Disease      Binary -0.0440
## bmi_bl                                                Contin. -0.0122
## mallampati_bl_1. 1                                     Binary  0.1437
## mallampati_bl_2. 2                                     Binary -0.0616
## mallampati_bl_3. 3+                                    Binary -0.1054
## smoking_bl_1. Current Smoker                           Binary -0.0067
## smoking_bl_2. Former Smoker                            Binary -0.0057
## smoking_bl_3. Never Smoker                             Binary  0.0127
## pain_yn_bl_1. Yes                                      Binary -0.1865
##
## Sample sizes
##     0. 5g Sugar 1. 0.5g Licorice
## All         117              118
```

- Standardized Difference > 0.10: Correlation with outcome

Unadjusted Analysis: t-Test

```
licorice_30_min_pain_t_test <-
  t.test(
    pacu_30_min_throat_pain ~ arm, data = licorice_gargle
  )
licorice_30_min_pain_t_test
##
##  Welch Two Sample t-test
##
## data:  pacu_30_min_throat_pain by arm
## t = 4.8035, df = 157.3, p-value = 3.608e-06
## alternative hypothesis: true difference in means between group 0. 5g Sugar and g
## 95 percent confidence interval:
##  0.4429931 1.0617225
## sample estimates:
##      mean in group 0. 5g Sugar mean in group 1. 0.5g Licorice
##                      1.0258621                      0.2735043
# Note: t-test is presenting [Control] (Reference) - [Treatment]
diff(licorice_30_min_pain_t_test$estimate)
## mean in group 1. 0.5g Licorice
##                     -0.7523578
-rev(licorice_30_min_pain_t_test$conf.int)
## [1] -1.0617225 -0.4429931
```

ANCOVA - Code

```
licorice_30_min_pain_glm_adjusted <-
  glm(
    pacu_30_min_throat_pain ~ arm +
      age_bl + gender + asa_physical_status_bl + bmi_bl + mallampati_bl,
    data = licorice_gargle,
    family = gaussian
  )

# Robust Huber-White SEs
licorice_30_min_pain_ancova_robust_vcov <-
  sandwich::vcovHC(licorice_30_min_pain_glm_adjusted, type = "HC0")

# CIs based on Robust SEs - lmtest::coeftest for Tests
licorice_30_min_pain_ancova_robust_ci <-
  lmtest::coefci(
    x = licorice_30_min_pain_glm_adjusted,
    vcov. = licorice_30_min_pain_ancova_robust_vcov
  )
```

ANCOVA - Results

```
##
## z test of coefficients:
##
##                                               Estimate Std. Error z value  Pr(>|z|)
## (Intercept)                                  1.3557577  0.5555588  2.4403   0.01467 *
## arm1. 0.5g Licorice                         -0.7219391  0.1510823 -4.7784 1.767e-06 ***
## age_bl                                      -0.0044495  0.0062754 -0.7090   0.47830
## gender1. Male                                0.2889391  0.1538593  1.8779   0.06039 .
## asa_physical_status_bl2. Mild Systemic Disease   0.2428688  0.2128941  1.1408   0.25395
## asa_physical_status_bl3. Severe Systemic Disease 0.3339201  0.2635809  1.2669   0.20521
## bmi_bl                                      -0.0264722  0.0192630 -1.3743   0.16936
## mallampati_bl2. 2                            0.2889037  0.1733967  1.6661   0.09568 .
## mallampati_bl3. 3+                           0.1828057  0.2619940  0.6977   0.48534
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##                                                    2.5 %       97.5 %
## (Intercept)                                   0.26688241  2.444633053
## arm1. 0.5g Licorice                          -1.01805508 -0.425823178
## age_bl                                       -0.01674907  0.007850052
## gender1. Male                                -0.01261948  0.590497703
## asa_physical_status_bl2. Mild Systemic Disease   -0.17439596  0.660133622
## asa_physical_status_bl3. Severe Systemic Disease -0.18268894  0.850529212
## bmi_bl                                       -0.06422693  0.011282550
## mallampati_bl2. 2                            -0.05094757  0.628754917
## mallampati_bl3. 3+                           -0.33069311  0.696304498
```

G-computation from GLM

```
# Step 2: Use working model to predict outcomes under each treatment
pred_30_min_pain_licorice <-
  predict(
    object = licorice_30_min_pain_glm_adjusted,
    newdata =
      within(
        data = licorice_gargle,
        expr = {arm = "1. 0.5g Licorice"}
      ),
    type = "response"
  )

pred_30_min_pain_sucrose <-
  predict(
    object = licorice_30_min_pain_glm_adjusted,
    newdata =
      within(
        data = licorice_gargle,
        expr = {arm = "0. 5g Sugar"}
      ),
    type = "response"
  )

# Step 3: Average predictions
mean_pred_30_min_pain_licorice <- mean(pred_30_min_pain_licorice)
mean_pred_30_min_pain_sucrose <- mean(pred_30_min_pain_sucrose)

# Step 4: Contrast Predictions
difference_licorice_sucrose <-
  mean_pred_30_min_pain_licorice - mean_pred_30_min_pain_sucrose
```

G-computation: drwls::standardization

```
licorice_30_min_pain_g_computation <-
  drwls::standardization(
    outcome_formula =
      pacu_30_min_throat_pain ~ tx + age_bl + gender +
      asa_physical_status_bl + bmi_bl + mallampati_bl,
    outcome_family = gaussian,
    treatment_column = "tx",
    estimand = "difference", # Compute Difference in Means
    data = licorice_gargle,
    se_method = "asymptotic" # "bootstrap"
  )
```

G-computation: `drwls::standardization`

```
##
## Estimand: Difference in Means
## Method: Marginal Standardization (G-Computation)
## Outcome Models: a gaussian GLM with an identity link
##
##
##                 Estimand   Estimate        SE       LCL        UCL         Z      p-value
## 1 E[Y|A=1] - E[Y|A=0] -0.7219391 0.1510823 -1.018055 -0.4258232 -4.778448 1.766534e-06
## 2            E[Y|A=1]  0.2887267        NA        NA         NA        NA           NA
## 3            E[Y|A=0]  1.0106658        NA        NA         NA        NA           NA
##
## Variance Estimation: Sandwich Estimator
## CI Method: Wald 95% CI
## Details:
##   Outcomes: (N = 233/235)
##   Outcome Model: pacu_30_min_throat_pain ~ tx + age_bl + gender + asa_physical_status_bl +      bmi_bl + ma
```

- Detailed but compact output: Estimand; Method; Working Model; Methods for SE + CI; Missing Data; Formulas
- Easy to extract high-level components (estimates, SE, CI, *p*-values) and low level components (models, predictions, variance adjustments)

ANCOVA vs. G-computation

```
diff(licorice_30_min_pain_t_test$estimate) # T-test
## mean in group 1. 0.5g Licorice
##                          -0.7523578
difference_licorice_sucrose # G-computation: Manual
## [1] -0.7219391
licorice_30_min_pain_g_computation$result$estimate[1] # G-computation
## [1] -0.7219391
coef(licorice_30_min_pain_glm_adjusted)["arm1. 0.5g Licorice"] # ANCOVA
## arm1. 0.5g Licorice
##           -0.7219391
```

- ANCOVA/G-Computation Match; Similar to t-test: Standard Errors smaller

```
licorice_30_min_pain_t_test$stderr # Unadjusted SE
## [1] 0.1566277
ate_i <- # ANOVA: Robust SE
  which(names(coef(licorice_30_min_pain_glm_adjusted)) == "arm1. 0.5g Licorice")
sqrt(licorice_30_min_pain_ancova_robust_vcov[ate_i, ate_i])
## [1] 0.1510823
licorice_30_min_pain_g_computation$result$se[1] # G-computation: Robust SE
## [1] 0.1510823
```

Computing Relative Efficiency

- How much more precise is adjusted analysis? $SE(\hat{\theta}) = \sqrt{var(\hat{\theta})}$
- Precision is the reciprocal of variance

```
var_unadj <- (licorice_30_min_pain_t_test$stderr)^2
var_adj <- (licorice_30_min_pain_g_computation$result$se[1])^2

prec_unadj <- 1/var_unadj
prec_adj <- 1/var_adj

prec_adj/prec_unadj # Improvement in Precision
## [1] 1.074756
1 - (var_adj/var_unadj) # Reduction in variance
## [1] 0.06955598
```

- This works for any type of analysis: getting CI more difficult

Missing Data

Taxonomy of Missing Data: (Rubin 1976)

- $R_i := 1$ if Outcome Y_i observed; 0 otherwise - Completness
- Missingness Mechanism: How does probability of being observed relate to the data at hand?
- Missing completely at random (MCAR): $Pr\{R|Y, A, X\} = Pr\{R\}$
  - Unrelated to baseline status, treatment assignment: **Testable**
- Missing at random (MAR): $Pr\{R|Y, A, X\} = Pr\{R|A, X\}$
  - Only related to what we observe: Least restrictive assumption that does not involve unobserved data
- Missing Not At Random (MNAR): Missingness depends on data that is not observed
  - Dropout: MAR/MCAR: Data can't distinguish - Sensitivity analysis

Covariate Adjustment & Missing Data

- Guidance on missing data: (Benkeser et al. 2020; Morris et al. 2022)
- Covariates: Single imputation or Multiple Imputation, but make sure imputation preserves $A \perp X$
- Complete case unadjusted analyses only valid if outcomes are MCAR.
- G-computation: outcomes are MCAR **or** model correctly specified
- Doubly robust methods: Model attrition process - Correct bias induced by outcomes which are missing at random.
  - Consistent estimate if outcomes are MAR and either outcome model or missingness model is correctly specified (Bang and Robins 2005).
  - Augmented Inverse Probability Weighted (AIPW) Estimator, Targeted Maximum Likelihood Estimator (TMLE), Doubly Robust Weighted Least Squares (DRWLS)

Doubly Robust Weighted Least Squares (Robins et al. 2007)

1. **Fit a missing data model** predicting complete (1) vs. missing (0) outcome: off-study observations: $E[R|A, X] = g_R^{-1}(X, A; \beta_R)$
   - Logistic Regression: Completeness indicator R on A, X
2. **Create a weight for each individual** that is the inverse of their predicted probability of being observed: $w_i = 1/\hat{R}_i = 1/g_R^{-1}(\hat{\beta}_{0(R)} + \hat{\beta}_{0(A)}A + \hat{\beta}_{X(R)}X)$
   - Give greater weight to complete cases
3. **Fit a weighted GLM for the outcome** using a canonical link among those with complete data: $E[Y|A, X, R = 1]$
4. **Predict** the outcome for each randomized individual i under each treatment assignment $a \in \{0, 1\}$, denoted $\hat{Y}_i^{(a)}$
5. **Average** these predictions over the sample: $\hat{\mu}^{(a)} = (1/n)\sum_{i=1}^{n} \hat{Y}_i^{(a)}$
6. **Contrast** the average predictions: Difference, Ratio

drwls::drwls - Code

```
licorice_30_min_pain_drwls <-
  drwls::drwls(
    outcome_formula = # Treatment-Stratified: List of Formulas
      list(
        pacu_30_min_throat_pain ~ age_bl + gender +
          asa_physical_status_bl + bmi_bl + mallampati_bl,
        pacu_30_min_throat_pain ~ age_bl + gender +
          asa_physical_status_bl + bmi_bl + mallampati_bl
      ),
    outcome_family = gaussian,
    treatment_column = "tx",
    missing_formula =  ~ tx + age_bl + bmi_bl + mallampati_bl,
    estimand = "difference", # Compute Difference in Means
    data = licorice_gargle,
    se_method = "asymptotic", # "bootstrap"
    # Use Tsiatis (2008) Small Sample Variance Correction
    variance_adjustment = variance_adjustment_tsiatis
  )
```

drwls::drwls - Results

```
##
## Estimand: Difference in Means
## Method: Doubly-Robust Weighted Least Squares
## Outcome Models: treatment-stratified gaussian GLMs with an identity link
## Censoring Model: a binomial GLM using a logit link
##
##              Estimand   Estimate        SE       LCL        UCL         Z      p-value
## 1 E[Y|A=1] - E[Y|A=0] -0.7141243 0.1547758 -1.017479 -0.4107694 -4.613929 3.951273e-06
## 2            E[Y|A=1]  0.2803998        NA        NA         NA        NA           NA
## 3            E[Y|A=0]  0.9945241        NA        NA         NA        NA           NA
##
## Variance Estimation: Sandwich Estimator with Tstiatis et al. (2008) adjustment
## CI Method: Wald 95% CI with Tstiatis et al. (2008) adjustment
## Details:
##   Outcomes: (N = 233/235)
##   Outcome Formula - A=1 (N = 117/118): pacu_30_min_throat_pain ~ age_bl + gender + asa_physical_status_bl +
##   Outcome Formula - A=0 (N = 116/117): pacu_30_min_throat_pain ~ age_bl + gender + asa_physical_status_bl +
##   Missing Formula: !is.na(pacu_30_min_throat_pain) ~ tx + age_bl + bmi_bl + mallampati_bl
```

drwls::drwls - Risk Ratio

```
licorice_30_min_any_pain_rr_drwls <-
  drwls::drwls(
    outcome_formula = # Treatment-Stratified: List of Formulas
      list(
        pacu_30_min_any_throat_pain ~ age_bl + gender +
          asa_physical_status_bl + bmi_bl + mallampati_bl,
        pacu_30_min_any_throat_pain ~ age_bl + gender +
          asa_physical_status_bl + bmi_bl + mallampati_bl
      ),
    outcome_family = gaussian,
    treatment_column = "tx",
    missing_formula =  ~ tx + age_bl + bmi_bl + mallampati_bl,
    estimand = "ratio", # Compute Relative Risk
    data = licorice_gargle,
    se_method = "bootstrap", # Delta method not yet implemented
    # Use Tsiatis (2008) Small Sample Variance Correction
    variance_adjustment = variance_adjustment_tsiatis
  )
```

drwls::drwls - Risk Ratio Results

```
##
## Estimand: Ratio of Means
## Method: Doubly-Robust Weighted Least Squares
## Outcome Models: treatment-stratified gaussian GLMs with an identity link
## Censoring Model: a binomial GLM using a logit link
##
##            Estimand  Estimate         SE       LCL       UCL         Z p-value
## 1 E[Y|A=1]/E[Y|A=0] 0.5525840 0.13341535 0.3061892 0.9099776 -3.353557      NA
## 2          E[Y|A=1] 0.1946247 0.03841988 0.1213197 0.2956318        NA      NA
## 3          E[Y|A=0] 0.3522083 0.04567178 0.2545035 0.4588395        NA      NA
##
## Variance Estimation: Non-parametric bootstrap using 10000 replicates with Tstiatis et al. (2008) adjustment
## CI Method: Bias Corrected and Accelerated (BCa) 95% CI with Tstiatis et al. (2008) adjustment
## Details:
##   Outcomes: (N = 233/235)
##   Outcome Formula - A=1 (N = 117/118): pacu_30_min_any_throat_pain ~ age_bl + gender + asa_physical_status_
##   Outcome Formula - A=0 (N = 116/117): pacu_30_min_any_throat_pain ~ age_bl + gender + asa_physical_status_
##   Missing Formula: !is.na(pacu_30_min_any_throat_pain) ~ tx + age_bl + bmi_bl +      mallampati_bl
```

Why use `drwls` Package?

- Continuous, Binary, Count outcomes: `outcome_family`
- Doubly Robust: `drwls::drwls()` - Formula for missingness model
  - MAR + either outcome or missingness model correctly specified
- Stratified Models/Treatment-covariate interactions: Improve precision (Ye et al. 2022)
- Impute missing covariates: Mean/Mode imputation; Can use `mice::mice`
- Bootstrap or Sandwich variance estimation
- Small sample size degree of freedom corrections (Tsiatis et al. 2008)

In Sum: Hands-On

- HTML file - Examples: Binary, Ordinal, Time-to-Event
- **R** has broadest range of methods; Room for improvement in usability; Software planning & validation
  - Syntax largely follows regression: Additional arguments
  - Off-the-shelf software; Macros; Dedicated packages
- On-line tutorials and data resources: Learning & Teaching
- Can use estimates of relative efficiency with historical data

FAQ

Learning Objectives: FAQ

- Identify common points of confusion about covariate adjustment and how to clarify these issues to the study team.
  - Role of Observed Balance in Precision Gain
  - Selecting covariates
  - Efficiency Guarantees
  - Missing covariates
  - Longitudinal or Clustered Data; Cluster Randomized Trials

FAQ

Questions About Covariate Adjustment

- Where to start? (Van Lancker, Bretz, and Dukes 2024; Bannick, Yi, and Ye 2026; U.S. Food and Drug Administration 2023; Benkeser et al. 2020)
- How to decide which variables to include?
  - Prognostic; Pre-specified *a priori*; Data from similar population
  - Data adaptive: Active research; Large trials; Crossfitting
- Should adjustment be guided by observed balance?
  - Change in Point estimate: Association & Balance
  - Efficiency: Association & *Variance in imbalance*: Still gain if balanced
- Can covariates be correlated? Is multicollinearity an issue?
  - Prediction not interpretation; Not testing coefficients
  - Correlated: Each covariate has incremental prognostic value

Questions About Covariate Adjustment

- Can covariate adjustment reduce precision?
  - Guarantees asymptotic; Not all methods have guarantees (Colantuoni and Rosenblum 2015); Weak/null association: increase variance
  - Corrections: CI Coverage; # of parameters (Tsiatis et al. 2008)
- Can covariate adjustment be beneficial in large trials?
  - Yes: *Absolute* imbalance decrease with N, *standardized differences* do not (Senn 1989). Imbalance is an issue for any size trial.
  - Larger trial: same precision gain = larger absolute reduction in N
- Can adjustment leverage post-randomization covariates?
  - Selection Bias; Missingness/Censoring: Plausibility; Specific Methods

Questions About Covariate Adjustment

- Can covariate adjustment play nicely with others?
  - ✓ Covariate Adaptive Randomization: Variance Estimation
  - ✓ Group Sequential Design: Orthogonalization (Van Lancker, Betz, and Rosenblum 2025)
- How should covariate adjustment be used in design calculations?
  - Gain: Magnitude, Precision, Transportability (Siegfried, Senn, and Hothorn 2022)
  - Rarely precisely known *a priori*: Transportability always a question
  - Conservative: Unadjusted; *Insurance* against other factors
  - Information Monitoring: Adaptive trial design - Reduce sample size (Mehta and Tsiatis 2001; Van Lancker, Betz, and Rosenblum 2025)

Questions About Covariate Adjustment

- Adjust for individual variables or prognostic scores? PROCOVA
  - PROCOVA = ANCOVA: prognostic modeling to create “super-covariate”
  - Assumes no treatment-covariate interactions; May be less efficient
  - Still faces issues of imprecision and transportability
- How to address missing data in covariates?
  - Methods uphold independence of treatment and covariates
  - Exclude covariates with lots of missing data, Mean/Mode imputation (Benkeser et al. 2020)
  - More complex imputation may improve efficiency: $A \perp\!\!\!\perp X_{\bar{S}}$; Subtleties: When to apply Rubin’s Rules? (Morris et al. 2022)
  - Doubly Robust methods (Bang and Robins 2005; Kang and Schafer 2007)

Questions About Covariate Adjustment

- Can covariate adjustment be used in longitudinal analyses? Yes - ANHECOVA can be generalized to MMRM framework (Wang and Du 2023)
- Can covariate adjustment be used in cluster randomized trials?
  - Yes, even designs with stratified randomization, minimization, and restricted randomization (Hemming et al. 2025; Li et al. 2015; Wang et al. 2026)
  - Addresses unique challenge: Post-randomization Selection Bias
  - Cluster RCTs: other issues & considerations (Kahan et al. 2022; Hemming and Taljaard 2023)

Questions About Covariate Adjustment

- Can you use machine learning for augmentation terms?
  - Yes: Bias/Variance Tradeoff; Data Adaptive Approaches
  - Large Trials & Non-linear Associations
- General Guidance on Approach: (Shao et al. 2026)
  - In general, simple models and selection strategies worked best
- How much precision can you gain?
  - Depends: Correlation between baseline and outcomes; Transportability of association to new trial (Siegfried, Senn, and Hothorn 2022)
  - Higher end: 20-25% with good prognostic scores

In Sum: FAQ in Practice

1. Choose covariates *a priori* - natural history: Correlated but contain *independent prognostic information* (prediction not interpretation): partial R^2. Low missingness - Imputation preserves $A \perp X$. Weak/noisy predictors can reduce precision.
2. Efficiency gains depends on the prognostic value and *the variance of the imbalance*, not the observed imbalance.
3. The difference between unadjusted and adjusted estimates depends on the prognostic value and the observed imbalance. Analyses shouldn't be guided by observed imbalance.
4. Precision gains still important in large studies: same gain means larger proportional reduction in N.
5. Bias-Variance tradeoff vs. Sample Size: Benefit of flexible models may not be realized at feasible sample sizes.
6. Best to design trials assuming no gain (imprecision, transportability): Use flexible study designs to reduce sample size.

References

Learning & Using R

- R For Data Science, 2nd Ed
- R Studio Education
- R Studio Cheatsheets
- R Studio Books
- Awesome R Learning Resources
- pharmaverse
- Hands-On Programming with R
- R Programming for Data Science
- R Markdown: The Definitive Guide
- Advanced R
- Quarto
- swirl
- Modern Statistics with R
- The R Warehouse
- Happy Git with R
- Posit Recipes
- Data Carpentry
- Reddit: r/Rlanguage and r/Rprogramming
- R Packages 2nd Ed
- Coursera Data Science Foundations in R
- CRAN Task Views

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